



Advanced Time Series Analysis and Forecasting

Seminar 0: Refresher and inference for dependent data



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CBB: Circular Block Bootstrap
- ▣ CEE: Central and Eastern Europe
- ▣ CLT: Central Limit Theorem
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ EWC: Equal-Weighted Cosine (estimator)
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GDP: Gross Domestic Product
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ LLSW: Lazarus, Lewis, Stock and Watson (2018)
- ▣ MA: Moving Average
- ▣ MBB: Moving-Block Bootstrap
- ▣ NW: Newey–West (standard errors)
- ▣ OLS: Ordinary Least Squares
- ▣ SE: Standard Error
- ▣ SPA: Superior Predictive Ability (test)
- ▣ TSA: Time Series Analysis

Today's question and route

- **Question:** when data are dependent, how wrong is the textbook standard error, and how do we fix it?
 - ▶ this is the first seminar of the master course and comes **before** Lecture 0: "Prerequisites for Today" gives every definition the tasks use
- **Route**
 - ▶ Part A: long-run variances, a Newey–West estimate by hand, a negative variance, block bootstraps by hand, multiple testing
 - ▶ Part B: the S&P 500 mean return; Romanian GDP growth; Monte Carlo of size; a replication of Granger and Newbold (1974); EUR/RON
 - ▶ Part C: data snooping on the BET; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1	long-run variance of AR(1) and MA(1); size of the naive test	Solved	–
A2, A3, A4	Newey–West by hand; a negative truncated estimate; why Bartlett is nonnegative	Solved, Proposed	A2
A5, A6	moving-block and circular bootstrap by hand	Solved, Proposed	A5
A7, A8	family-wise error, Bonferroni and Holm	Solved, Proposed	A7
B1, B2	HAC inference for a mean: S&P 500, Romanian GDP	Solved, Proposed	B1
B3, B4	Monte Carlo of size: AR(1) and GARCH(1,1)	Solved, Proposed	B3
B5, B6	replication of Granger and Newbold (1974); block bootstrap of EUR/RON	Solved, Proposed	B5, B1
C1, C2	data snooping on the BET; what is wrong in an AI answer?	Proposed	A7, B1–B6

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Learning outcomes

- ▣ Compute a long-run variance from a model and from data, by hand and in code
- ▣ Explain why the Bartlett kernel gives a nonnegative variance and the truncated kernel may not
- ▣ Run a block bootstrap and choose its block length
- ▣ Measure the actual size of a test by Monte Carlo and report its Monte Carlo error
- ▣ Replicate a published simulation result and explain the gap

Data used

Series	Source	Frequency	Period
S&P 500, BET	EODHD	daily log returns, %	2000–2026
real GDP, Romania	Eurostat (namq_10_gdp)	quarterly, q/q growth, %	2000Q1–2026Q2
EUR/RON	BNR reference rate	daily log change, %	2005–2026
simulated AR(1), GARCH(1,1), random walks	–	–	seed 2026

- In the notebook: `acov_all`, `lrsv`, `andrews_bw`, `ewc_lrsv`, `fixed_b_table`, `politis_white`, `mbb_means`, `sb_indices`; no account or key is needed

Prerequisites for Today (1/6): the long-run variance

- x_t , $t = 1, \dots, T$: a weakly stationary series with mean μ and sample mean $\bar{x} = T^{-1} \sum_t x_t$
 - ▶ $\gamma_j = \text{Cov}(x_t, x_{t-j})$: the autocovariance at lag j ; γ_0 is the variance; $\rho_j = \gamma_j/\gamma_0$: the autocorrelation
- The variance of the mean adds the covariances of all pairs of observations:
 - ▶ $\text{Var}(\sqrt{T}\bar{x}) = \sum_{|j| < T} (1 - |j|/T) \gamma_j \rightarrow \Omega = \gamma_0 + 2 \sum_{j \geq 1} \gamma_j$
 - ▶ $1 - |j|/T$: the share of pairs at distance j ; as T grows, every weight tends to 1
- Ω is the **long-run variance**
 - ▶ the i.i.d. formula keeps only γ_0 ; $\Omega/\gamma_0 > 1$ means the naive standard error is too small

Prerequisites for Today (2/6): models, CLT and size

- ARMA (TSA, Chapter 2): $\phi(L)x_t = \theta(L)\varepsilon_t$, with L the lag operator and ε_t white noise with variance σ^2
 - ▶ $\Omega = \sigma^2\theta(1)^2/\phi(1)^2$, where $\phi(1) = 1 - \sum_i \phi_i$ and $\theta(1) = 1 + \sum_i \theta_i$
 - ▶ AR(1) $x_t = \phi x_{t-1} + \varepsilon_t$: $\gamma_j = \phi^{|j|}\sigma^2/(1 - \phi^2)$, $\Omega = \sigma^2/(1 - \phi)^2$, so $\Omega/\gamma_0 = (1 + \phi)/(1 - \phi)$
- CLT: $\sqrt{T}(\bar{x} - \mu) \rightarrow_d N(0, \Omega)$ ($\rightarrow_d$: convergence in distribution)
 - ▶ holds under mixing and moment conditions, or for martingale differences, where $\Omega = \gamma_0$
- Size of a test: its rejection rate under a true H_0 ; nominal level 5%
 - ▶ the naive t uses $\hat{\gamma}_0$; under AR(1) it rejects a true H_0 with probability $2[1 - \Phi(1.96\sqrt{\gamma_0/\Omega})]$
 - ▶ Φ : the distribution function of the standard Normal distribution

Prerequisites for Today (3/6): HAC estimators

- Sample autocovariance (divisor T):
 - ▶ $\hat{\gamma}_j = T^{-1} \sum_{t=j+1}^T (x_t - \bar{x})(x_{t-j} - \bar{x})$
- Kernel estimator of Ω : a weighted sum of sample autocovariances
 - ▶ $\hat{\Omega} = \hat{\gamma}_0 + 2 \sum_{j \geq 1} k(j/S) \hat{\gamma}_j$
 - ▶ $k(\cdot)$: the kernel (weight function, $k(0) = 1$); S : the bandwidth; HAC: heteroskedasticity and autocorrelation consistent
- **Newey–West** [Newey and West \(1987\)](#): Bartlett weights $1 - j/(L + 1)$ for $j = 1, \dots, L$, i.e. $S = L + 1$
 - ▶ L : the number of lags included; the weights fall linearly to $1/(L + 1)$
 - ▶ **truncated** kernel: weight 1 for every lag up to L , 0 afterwards

Prerequisites for Today (4/6): bandwidths and critical values

- Rule of thumb **Newey and West (1994)**: $L = \lfloor 4(T/100)^{2/9} \rfloor$, with $\lfloor \cdot \rfloor$ the integer part
 - ▶ **Andrews Andrews (1991)**: $S = 1.1447(\hat{\alpha}T)^{1/3}$, $\hat{\alpha} = 4\hat{\rho}^2/[(1 - \hat{\rho})^2(1 + \hat{\rho})^2]$
 - ▶ $\hat{\rho}$: the estimated AR(1) coefficient of the series; more persistence gives a larger S
- **LLSW Lazarus et al. (2018)**: NW with $S = 1.3\sqrt{T}$ and **fixed- b** critical values **Kiefer and Vogelsang (2005)**
 - ▶ $b = S/T$; the fixed- b critical values are larger than 1.96 and grow with b , because they account for the noise in $\hat{\Omega}$
- **EWC** (equal-weighted cosine): an average of ν squared cosine projections
 - ▶ $\hat{\Omega} = \nu^{-1} \sum_{j=1}^{\nu} \hat{\Lambda}_j^2$, $\hat{\Lambda}_j = \sqrt{2/T} \sum_t \cos[\pi j(t - 1/2)/T](x_t - \bar{x})$, $\nu = 0.4T^{2/3}$
 - ▶ ν : the number of cosines and the degrees of freedom: critical values from Student t_ν

Prerequisites for Today (5/6): bootstrap and Monte Carlo

- **Moving-block bootstrap** (MBB) [Künsch \(1989\)](#): draw $\lceil T/l \rceil$ overlapping blocks of length l with replacement and concatenate
 - ▶ $\lceil \cdot \rceil$: rounding up; a star marks a bootstrap quantity: $\bar{x}^*$, E^* , Var^* (mean, expectation and variance over resamples)
 - ▶ **circular** (CBB): blocks wrap around the end; **stationary** [Politis and Romano \(1994\)](#): geometric block lengths with mean $1/p$
 - ▶ block length: Politis–White rule [Politis and White \(2004\)](#), [Patton, Politis and White \(2009\)](#); the i.i.d. bootstrap is the case $l = 1$
 - ▶ basic 95% interval: $[2\bar{x} - q_{0.975}^*, 2\bar{x} - q_{0.025}^*]$, with q_a^* the a -quantile of $\bar{x}^*$
- **Wild bootstrap**: multiply each residual by an independent random sign
 - ▶ keeps heteroskedasticity, destroys autocorrelation
- **Monte Carlo**: simulate R samples under H_0 and count the rejections
 - ▶ p : the rejection rate; its Monte Carlo standard error is $\sqrt{p(1-p)/R}$

Prerequisites for Today (6/6): multiple testing

- K tests, each at level α
 - ▶ **FWER** (family-wise error rate) = $P(\text{at least one false rejection})$
 - ▶ for K independent tests of true hypotheses: $\text{FWER} = 1 - (1 - \alpha)^K$
- **Bonferroni**: test each hypothesis at α/K
 - ▶ controls the FWER under any dependence between the tests
- **Holm** (step-down): sort the p -values $p_{(1)} \leq \dots \leq p_{(K)}$
 - ▶ reject while $p_{(i)} \leq \alpha/(K - i + 1)$ and stop at the first failure
 - ▶ the first threshold is the Bonferroni one; the later ones are less strict

A1: long-run variances and the naive test [Solved]

Task

- ▣ AR(1) $x_t = 0.6x_{t-1} + \varepsilon_t$ and MA(1) $y_t = \varepsilon_t + 0.5\varepsilon_{t-1}$, $\sigma^2 = 1$.
- ▣ 1. Compute γ_0 , Ω and Ω/γ_0 for both processes.
- ▣ 2. Compute the asymptotic size of the naive 5% t -test for the AR(1) mean.
- ▣ 3. Compute Ω/γ_0 for $\phi = -0.6$ and say what happens to the naive test.
- ▣ Report: two ratios, one size, one sentence.

Solution

- ▣ 1. AR(1): $\gamma_0 = 1/(1 - 0.36) = 1.5625$, $\Omega = 1/0.4^2 = 6.25$, ratio $1.6/0.4 = 4.00$
- ▣ MA(1): $\gamma_0 = 1 + \theta^2 = 1.25$, $\gamma_1 = \theta = 0.50$, $\Omega = (1 + \theta)^2 = 2.25$, ratio 1.80
- ▣ 2. The naive t is $N(0, \Omega/\gamma_0) = N(0, 4)$: size $= 2[1 - \Phi(1.96/2)] = 32.7\%$
- ▣ 3. $\phi = -0.6$: ratio $0.4/1.6 = 0.25$; the naive test almost never rejects: it is conservative and has little power

A2: Newey–West by hand [Solved]

Task

- ▣ An illustrative series of $T = 10$ values:
(2.1, 1.4, 0.3, -0.5, 0.2, 1.1, 1.8, 0.9, -0.2, 0.4).
- ▣ 1. Compute $\bar{x}$, $\hat{\gamma}_0$, $\hat{\gamma}_1$, $\hat{\gamma}_2$ with divisor T .
- ▣ 2. Compute the Newey–West $\hat{\Omega}$ with $L = 2$ (the rule of thumb for $T = 10$).
- ▣ 3. Compare the naive and the HAC t -statistics of $H_0 : \mu = 0$.
- ▣ Report: three autocovariances, $\hat{\Omega}$, two t -statistics.

Solution

- ▣ 1. $\bar{x} = 0.7500$; $\hat{\gamma}_0 = 0.6585$, $\hat{\gamma}_1 = 0.2357$, $\hat{\gamma}_2 = -0.3185$
- ▣ 2. Weights $1 - 1/3 = 2/3$ and $1 - 2/3 = 1/3$:
 $\hat{\Omega} = \hat{\gamma}_0 + 2(\frac{2}{3}\hat{\gamma}_1 + \frac{1}{3}\hat{\gamma}_2) = 0.7605$
- ▣ 3. SE naive $\sqrt{\hat{\gamma}_0/10} = 0.257$, $t = 2.923$; SE HAC
 $\sqrt{\hat{\Omega}/10} = 0.276$, $t = 2.720$
- ▣ The positive $\hat{\gamma}_1$ and the negative $\hat{\gamma}_2$ partly cancel; with $T = 10$ the estimate is very noisy (fixed- b would raise the critical value)

A3: a negative variance [Proposed]

Solution discussed in class

Task

- ▣ The series $(1, -1, 1, -1, 1, -1, 1, -1, 2, -2)$. Model: A2.
- ▣ 1. Compute $\hat{\gamma}_0, \hat{\gamma}_1, \hat{\gamma}_2$.
- ▣ 2. Compute the truncated-kernel estimate with $L = 1$ and the Newey–West estimate with $L = 1$.
- ▣ 3. Compute the Newey–West estimate with $L = 2$.
- ▣ Report: three autocovariances, three variance estimates, one sentence on the sign.

A4: why the Bartlett kernel is nonnegative [Proposed]

Solution discussed in class

Task

- ▣ Demeaned data $u_1, \dots, u_T$; Newey–West with L lags. Model: A2.
- ▣ 1. Let $s_t = \sum_{i=0}^L u_{t-i}$ (with $u_t = 0$ outside the sample). Show that $\frac{1}{(L+1)T} \sum_{t=1}^{T+L} s_t^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^L \left(1 - \frac{j}{L+1}\right) \hat{\gamma}_j$.
- ▣ 2. Conclude that $\hat{\Omega}_{NW} \geq 0$ for every sample.
- ▣ 3. Explain why no such representation exists for the truncated kernel.
- ▣ Report: the derivation and one sentence.

A5: the moving-block bootstrap by hand [Solved]

Task

- ▣ Series $x = (1, 3, 2, 6, 4, 2)$, block length $l = 2$, so $k = 3$ blocks per resample.
- ▣ 1. List the moving blocks and their means.
- ▣ 2. Compute $E^* \bar{x}^*$ and compare it with $\bar{x}$.
- ▣ 3. Compute $\text{Var}^*(\bar{x}^*)$ and compare it with $\hat{\gamma}_0/T$ and with the Bartlett estimate $\hat{\Omega}(S=2)/T$.
- ▣ Report: five block means, two expectations, three variances.

Solution

- ▣ 1. Blocks $(1, 3), (3, 2), (2, 6), (6, 4), (4, 2)$; means 2.0, 2.5, 4.0, 5.0, 3.0
- ▣ 2. $\bar{x}^*$ is the average of 3 block means drawn uniformly: $E^* \bar{x}^* = 3.3000 \neq \bar{x} = 3.0000$: the end values 1 and 2 enter only one block each (edge bias)
- ▣ 3. $\text{Var}^*(\bar{x}^*) = \text{Var}^*(\text{block mean})/3 = 0.3867$; naive $\hat{\gamma}_0/6 = 0.4444$; Bartlett 0.4167
- ▣ With $T = 6$ the three numbers differ; asymptotically MBB with block l and Bartlett with $S = l$ coincide [Künsch \(1989\)](#)

A6: the circular block bootstrap by hand [Proposed]

Solution discussed in class

Task

- ▣ The same series and $l = 2$; wrap the series on a circle. Model: A5.
- ▣ 1. List the six circular blocks and their means.
- ▣ 2. Show that $E^* \bar{x}^* = \bar{x}$ and explain why.
- ▣ 3. Compute $\text{Var}^*(\bar{x}^*)$.
- ▣ Report: six block means, one expectation, one variance, one sentence.

A7: the family-wise error [Solved]

Task

- ▣ A researcher tests 20 independent, useless predictors of inflation, each at 5%.
- ▣ 1. Compute the probability of at least one rejection (FWER).
- ▣ 2. Compute the expected number of false rejections.
- ▣ 3. Give the Bonferroni level and the resulting FWER.
- ▣ Report: two probabilities, one count, one level.

Solution

- ▣ 1. $\text{FWER} = 1 - 0.95^{20} = 64.2\%$
- ▣ 2. $E[\text{\#false rejections}] = 20 \cdot 0.05 = 1$
- ▣ 3. Bonferroni: $0.05/20 = 0.0025$; $\text{FWER} = 1 - (1 - 0.0025)^{20} = 4.88\% \leq 5\%$
- ▣ Bonferroni holds under any dependence (union bound); with positively correlated tests it is conservative

A8: Holm step-down [Proposed]

Solution discussed in class

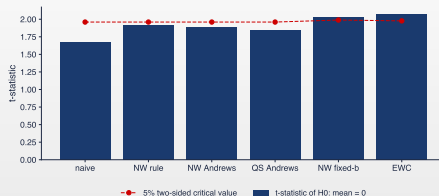
Task

- ▣ Six sorted p -values: 0.001, 0.008, 0.012, 0.030, 0.041, 0.240; FWER target 5%. Model: A7.
- ▣ 1. Write the six Holm thresholds $0.05/(6 - i + 1)$.
- ▣ 2. Count the rejections with Holm, with Bonferroni and with no correction.
- ▣ 3. With 50 tests at 1%, give the FWER of uncorrected tests and the Bonferroni level.
- ▣ Report: six thresholds, three counts, two numbers.

B1: the mean return of the S&P 500 [Solved]

- **Question:** is the mean daily log return of the S&P 500 since 2000 significantly positive, and does the answer depend on the standard error?
- **Inputs:** S&P 500 daily log returns in %, 2000–2026 ($T = 6\,714$)
- **Tasks**
 1. Compute the mean and $\hat{\rho}_1$.
 2. Compute five standard errors: naive, Newey–West rule of thumb, Newey–West with the Andrews bandwidth, NW with $S = 1.3\sqrt{T}$ (fixed- b), EWC.
 3. Compute each t -statistic for $H_0 : \mu = 0$ and compare it with its critical value.
 4. Interpretation: why are the robust standard errors smaller than the naive one here?
- **Report:** one mean, one autocorrelation, five standard errors with their t -statistics, one sentence
- **Notebook:** section B1

B1: solution [Solved]



- $\bar{r} = 0.0247\%$ a day (6.23% a year), $\hat{\rho}_1 = -0.099$; SE: naive 0.0148, NW rule ($S = 11$) 0.0129, NW Andrews ($S = 7.4$) 0.0131, fixed- b ($S = 106.5$) 0.0122, EWC ($\nu = 142$) 0.0119
- t : naive 1.67, NW 1.91, Andrews 1.88 (all against 1.96); fixed- b 2.03 against 1.99; EWC 2.07 against 1.98: only the last two reject, barely
- Interpretation: the negative first-order autocorrelation makes $\Omega < \gamma_0$; daily index returns mean-revert slightly, so the naive test is conservative; the conclusion stays fragile

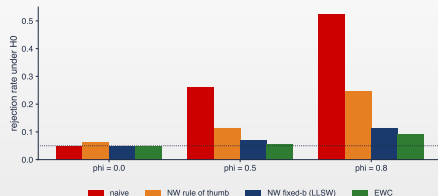
B2: the mean growth of Romanian GDP [Proposed]

- ▣ **Question:** how precisely do we know the average quarterly growth of Romanian real GDP since 2000?
- ▣ **Inputs:** Eurostat real GDP, seasonally adjusted, q/q log growth in %, 2000Q1–2026Q2 ($T = 106$); model: B1
- ▣ **Tasks**
 1. Compute the mean, $\hat{\rho}_1$ and the 95% intervals with the five methods of B1.
 2. Add a stationary-bootstrap percentile interval with the Politis–White block length (4999 resamples).
 3. Repeat the main estimates without the four quarters of 2020.
 4. Interpretation: what matters more for this mean, dependence or the 2020 quarters?
- ▣ **Report:** one mean, six intervals, one block length, one comparison, one sentence
- ▣ **Notebook:** section B2

B3: Monte Carlo of size under AR(1) [Solved]

- ▣ **Question:** with a sample as long as the Romanian GDP series, which test of a mean keeps its 5% size when the data are autocorrelated?
- ▣ **Inputs:** simulated AR(1), $\phi \in \{0, 0.5, 0.8\}$, $T = 106$, 4 000 replications, seed 2026
- ▣ **Tasks**
 1. Simulate the AR(1) paths with 200 burn-in values and $N(0, 1)$ innovations.
 2. For each path test $H_0 : \mu = 0$ at 5% with the naive test, NW rule of thumb, NW fixed- b and EWC.
 3. Report the rejection rates and their Monte Carlo standard errors.
 4. Interpretation: which test would you use for the mean of a series with $\hat{\rho}_1 = 0.5$ and $T \approx 100$?
- ▣ **Report:** a 3×4 table of rejection rates, one Monte Carlo error, one sentence
- ▣ **Notebook:** section B3

B3: solution [Solved]



- $\phi = 0$: naive 5.0%, NW 6.2%, fixed-b 4.8%, EWC 4.7%; $\phi = 0.5$: 26.3%, 11.4%, 6.9%, 5.4%; $\phi = 0.8$: 52.3%, 24.8%, 11.2%, 9.1%
- Monte Carlo standard error: $\sqrt{0.05 \cdot 0.95 / 4000} \approx 0.34$ pp at 5%, about 0.8 pp at 50%
- Interpretation: at $\phi = 0.5$ and $T \approx 100$ use EWC or NW fixed-b; the naive test and NW with the rule of thumb reject far too often, and the gap grows with ϕ

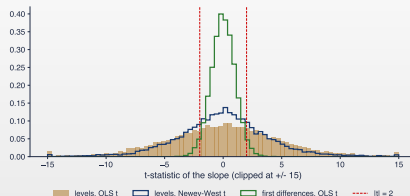
B4: Monte Carlo of size under GARCH(1,1) [Proposed]

- **Question:** does conditional heteroskedasticity alone distort a test on the mean of returns, and a test on the mean of squared returns?
- **Inputs:** simulated GARCH(1,1) $r_t = \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$ (σ_t^2 : conditional variance; z_t : $N(0, 1)$ innovations), $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$, $T = 1\,000$, 4 000 replications; model: B3
- **Tasks**
 1. Check that the unconditional variance is 1 and that the fourth moment exists ($3\alpha^2 + 2\alpha\beta + \beta^2 < 1$).
 2. Test $H_0 : Er_t = 0$ and $H_0 : Er_t^2 = 1$ at 5% with the four tests of B3.
 3. Compute ρ_1 of r_t^2 from the GARCH formula $\alpha(1 - \alpha\beta - \beta^2)/(1 - 2\alpha\beta - \beta^2)$.
 4. Interpretation: why is the first test fine and the second one not?
- **Report:** one moment condition, eight rejection rates, one autocorrelation, one sentence
- **Notebook:** section B4

B5: replicating Granger and Newbold (1974) [Solved]

- **Question:** how often does a regression of one random walk on an independent one look significant, and can HAC standard errors fix it?
- **Inputs:** Granger and Newbold (1974): $T = 50$, two independent Gaussian random walks, 100 regressions; published result: $|t| > 2$ in 76 of 100 cases
- **Tasks**
 1. Replicate the design: simulate y_t and x_t , regress y_t on $1, x_t$, record the OLS t of the slope; use first 100, then 10 000 replications.
 2. Recompute the share of $|t| > 2$ with Newey–West standard errors (rule of thumb) and with NW fixed- b .
 3. Repeat the regression in first differences.
 4. Interpretation: is the published 76% consistent with your replication?
- **Report:** four rejection shares, one Monte Carlo error, one sentence
- **Notebook:** section B5

B5: solution [Solved]



- First 100 replications: 68.0%; 10 000 replications: OLS 65.9%, Newey–West 54.2%, fixed- b 41.4% (critical value 2.48); first differences: 5.1%
- With 100 replications the Monte Carlo standard error is about 4.7 pp: 76% lies about two standard errors above our long-run rate, so the published number is a high draw, not a different phenomenon
- Interpretation: the result replicates in substance; HAC corrections reduce but do not remove the problem, because the t -statistic diverges Phillips (1986); differencing restores the 5% size

B6: block bootstrap of EUR/RON [Proposed]

- ▣ **Question:** has the leu depreciated significantly against the euro since 2005, on a daily mean basis?
- ▣ **Inputs:** EUR/RON BNR reference rate, daily log changes in %, 5 July 2005 to 18 September 2026 ($T = 5\,352$); model: B1
- ▣ **Tasks**
 1. Compute the mean, $\hat{\rho}_1$ and the Politis–White block lengths for the stationary and circular bootstrap.
 2. Compute bootstrap standard errors and 95% basic intervals with the i.i.d., moving-block, circular and stationary bootstrap (1999 resamples).
 3. Convert the mean into an average annual depreciation and compare it with the change of the rate between the first and the last day.
 4. Interpretation: why does the block bootstrap change so little here?
- ▣ **Report:** one mean, two block lengths, four intervals, one annual rate, one sentence
- ▣ **Notebook:** section B6

C1: data snooping on the BET [Proposed]

- **Question:** after adding momentum rules to the 50 moving-average rules of the lecture, does any rule beat buy-and-hold on the BET in 2013–2026?
- **Inputs:** BET daily closes (EODHD); rules: long if the close is above its n -day moving average ($n = 5, \dots, 250$) or if the n -day return is positive ($n = 5, 10, 20, 60, 120, 250$); model: A7, B1
- **Tasks**
 1. Write down, before computing, the rule universe, the sample and the benchmark (a pre-analysis plan).
 2. Compute the excess return of each rule over buy-and-hold and the naive t of the best rule.
 3. Run the Reality Check [White \(2000\)](#) with the stationary bootstrap (mean block 10, 999 resamples).
 4. Interpretation: how would transaction costs of 0.1% per trade change the conclusion?
- **Report:** a dated plan, one t -statistic, one Reality Check p -value, one sentence; possible project seed
- Notebook: section C1

C2: critique of an AI answer

- A student asked an AI assistant: "How do I get correct standard errors for the mean of daily squared returns?" The answer contained six statements:
 - ▶ (a) "Newey–West standard errors are always larger than OLS standard errors."
 - ▶ (b) "For maximum robustness use $L = T - 1$ lags."
 - ▶ (c) "The wild bootstrap handles the autocorrelation of squared returns."
 - ▶ (d) "With $T = 6700$ observations the normal critical value 1.96 is always accurate."
 - ▶ (e) "The truncated kernel is better because it does not down-weight any lag."
 - ▶ (f) "The stationary bootstrap of Politis and Romano (1994) produces a stationary resampled series."
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and the exercise of today that shows it.
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- ▣ The precision of a mean is set by $\Omega = \gamma_0 + 2 \sum_j \gamma_j$; positive dependence inflates it, negative dependence deflates it
- ▣ Bartlett (Newey–West) is a sum of squares and never negative; the truncated kernel can be
- ▣ Block bootstraps resample dependence; the circular version removes the edge bias
- ▣ A Monte Carlo of size is the honest check of any test: report R , the seed and the Monte Carlo error
- ▣ A replication is successful when the gap is explained, not when the digits match

After the seminar

- ▣ Lecture 0 develops each topic of today: ergodicity and mixing, the HAC estimators, fixed- b , the block bootstraps, data snooping and reproducible research
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 can grow into a team project: data snooping in CEE technical-trading or macro-forecasting studies
- ▣ Reading: [Newey and West \(1987\)](#); [Lazarus et al. \(2018\)](#); [Künsch \(1989\)](#); [Granger and Newbold \(1974\)](#)
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

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